

## LECTURE POINTS

### **Climate Change & Environmental Degradation**

#### **1. Introduction**

- Climate change and environmental degradation are interconnected global crises affecting human survival, ecosystems, and global politics.
- These issues are now part of high politics (security, economy, diplomacy), not just environmental concerns.
- Scientific consensus (especially by the Intergovernmental Panel on Climate Change (IPCC)) confirms that anthropogenic (human-caused) activities are the dominant factor.
- Climate change is now considered a “threat multiplier” because it worsens existing issues like poverty, conflict, and inequality.

**Example:** Climate stress can intensify political instability in already fragile states like Somalia.

#### **2. Meaning of Key Terms**

##### **Climate Change**

- Long-term alteration in global or regional climate patterns (temperature, precipitation, wind systems).
- Includes both:
  - Global warming (rise in average temperature)
  - Climate variability (extreme weather changes)
- Scientific basis: Linked to the Greenhouse Effect (heat trapped in atmosphere).

**Advanced Example:** Increase in CO<sub>2</sub> concentration from ~280 ppm (pre-industrial) to over 420 ppm today.

##### **Environmental Degradation**

- Broad concept covering decline in environmental quality due to human and natural factors.
- Includes:
  - Air and water pollution
  - Land degradation
  - Deforestation
  - Loss of biodiversity

**Analytical Point:** Environmental degradation reduces carrying capacity of ecosystems (ability to sustain life).

#### **3. Present Scenario**

- Climate change is now irreversible in the short term due to accumulated emissions.

- Global temperature rise is uneven → developing countries suffer more despite contributing less.

### **Additional Key Trends:**

- Ocean Acidification (due to CO<sub>2</sub> absorption) → harms marine life
- Heatwaves increasing in frequency and intensity
- Arctic ice shrinking, opening new geopolitical competition routes

**Example:** Melting Arctic ice is creating new trade routes and geopolitical tensions.

## **4. Causes**

### **A. Greenhouse Effect (Scientific Mechanism)**

- Solar radiation enters Earth → absorbed → re-radiated as heat
- Greenhouse gases trap this heat → warming effect

**Important Concept:** Without greenhouse gases, Earth would be too cold—but excess causes warming.

### **B. Human Activities**

#### **1. Deforestation**

- Forests act as carbon sinks (absorb CO<sub>2</sub>)
- Their removal increases carbon concentration

**Advanced Example:** Amazon rainforest turning from carbon sink to carbon source

#### **2. Industrialization**

- Fossil fuels (coal, oil, gas) → major emission sources
- Leads to industrial capitalism vs environmental sustainability debate

#### **3. Urbanization**

- Creates urban heat islands (cities hotter than rural areas)
- Increases energy demand and pollution

#### **4. Agriculture**

- Methane from livestock
- Nitrous oxide from fertilizers

**Example:** Rice fields produce methane emissions

#### **5. Overconsumption**

- Consumer culture increases waste and energy use

**Example:** Fast fashion industry contributing to pollution

## **5. Climate Change as a Security Threat**

**Concept: “Threat Multiplier”**

Climate change does not create conflict alone, but intensifies existing tensions.

### **A. Migration & Displacement**

- Leads to climate refugees (not legally recognized yet)

**Critical Insight:** Migration can strain urban areas and increase unemployment

### **B. Resource Wars**

- Water and land scarcity → interstate and intrastate conflicts

**Example:** Nile River dispute (Egypt–Ethiopia)

### **C. Water Scarcity**

- Glacial melt initially increases water → later causes shortages

**Example:** Indus River system dependency in Pakistan

### **D. Food Security**

- Climate change disrupts:
  - Growing seasons
  - Crop yields
  - Livestock health

**Advanced Point:** Food insecurity can lead to political unrest (Arab Spring link)

### **E. Economic Impact**

- Losses in:
  - Agriculture
  - Infrastructure
  - Insurance systems

**Example:** Developing countries spend more on recovery than development

### **F. Health Risks**

- Heat strokes
- Vector-borne diseases (malaria, dengue)

**Example:** Rising dengue cases in South Asia

## **6. Environmental Degradation as a Security Threat**

### **A. Ecosystem Collapse**

- When ecosystems fail, entire human systems collapse

Example: Fisheries collapse → unemployment → migration

### **B. Desertification**

- Land becomes unusable → rural poverty

Example: Thar Desert expansion in Pakistan

### **C. Biodiversity Loss**

- Loss of species disrupts food chains and ecological balance

Example: Decline in pollinators affects agriculture

### **D. Political Instability**

- Weak states cannot manage environmental stress

Example: Haiti's deforestation leading to social unrest

## **7. Major Impacts**

### **1. Sea-Level Rise**

- Salinization of agricultural land
- Threat to coastal economies

### **2. Extreme Weather Events**

- More frequent cyclones, floods, droughts

Example: Pakistan floods 2022 (economic + humanitarian crisis)

### **3. Infrastructure Damage**

- Energy systems, transport, housing destroyed

### **4. Agricultural Crisis**

- Crop failures → inflation → poverty

### **5. Global Inequality**

- Developed vs developing country divide increases

## **8. Solutions**

### **Sustainable Practices**

- Renewable energy (solar, wind, hydro)
- Energy efficiency

### **Global Cooperation**

- Climate agreements (Paris Agreement)
- Technology transfer to developing countries

### **Policy Measures**

- Carbon taxes
- Environmental regulations
- Climate adaptation strategies

### **Role of Society**

- Public awareness
- Behavioral change (reduce waste, conserve energy)

### **Climate Justice**

- Developed countries responsible for most emissions must support developing countries

## **Conclusion**

- Climate change is not just an environmental issue—it is a global political, economic, and security crisis.
- Without urgent action, it may lead to:
  - State failure
  - Global conflicts
  - Humanitarian disasters

Final Insight: Sustainable development is the only path forward, balancing economic growth and environmental protection.

# **Sustainable Development Goals (SDGs) in Pakistan**

## **1. Introduction**

- The Sustainable Development Goals (SDGs) are a global development agenda adopted in 2015 (Agenda 2030).
- They include 17 goals and 169 targets aimed at solving global issues.
- Focus on balance between:
  - Economic growth
  - Social inclusion
  - Environmental protection
- Core principle: “Leave No One Behind”
- SDGs go beyond MDGs by emphasizing:
  - Sustainability
  - Inclusiveness
  - Root causes of problems

**Insight:** SDGs are a global governance framework, not just development goals.

## **2. What are SDGs?**

- A comprehensive framework addressing:
  - Poverty, health, education
  - Inequality, climate change
  - Peace and justice

### **Key Features:**

- Interconnected
- Long-term (till 2030)
- Measurable

## **3. Key Features of SDGs**

- **Universal** → for all countries
- **Comprehensive** → covers all sectors
- **Integrated** → goals are linked
- **Inclusive** → focus on vulnerable groups
- **Partnership-based** → global cooperation

## **4. Importance of SDGs**

- Provide a roadmap for development
- Help in policy-making and planning
- Promote:
  - Economic growth
  - Social justice
  - Environmental sustainability

## **5. Pakistan's SDG Categorization**

Pakistan divides SDGs into three categories based on priority and resources.

### **Category 1: High Priority Goals**

These are most urgent and essential for survival and stability.

#### **Key Goals:**

- **SDG 2 (Zero Hunger)** → Food security
- **SDG 3 (Health)** → Healthcare access
- **SDG 4 (Education)** → Literacy & schooling
- **SDG 6 (Water)** → Clean drinking water
- **SDG 7 (Energy)** → Affordable energy
- **SDG 8 (Economy)** → Jobs & growth

#### **Analysis:**

- Focus on basic human needs
- Directly linked to poverty reduction and stability

### **Category 2: Medium Priority Goals**

These support long-term development and social progress.

#### **Key Goals:**

- SDG 1 (No Poverty)
- SDG 5 (Gender Equality)
- SDG 9 (Industry & Innovation)
- SDG 10 (Reduced Inequality)
- SDG 11 (Sustainable Cities)
- SDG 17 (Partnerships)

#### **Analysis:**

- Strengthen economic and social structures
- Improve quality of life and inclusiveness

### **Category 3: Lower Priority (Long-Term Goals)**

These are important but less urgent due to resource constraints.

#### **Key Goals:**

- SDG 13 (Climate Action)
- SDG 14 (Life Below Water)
- SDG 15 (Life on Land)
- SDG 12 (Responsible Consumption)

#### **Analysis:**

- Focus on environmental sustainability
- Neglecting them can harm long-term development

*All categories are interconnected—ignoring one can affect others.*

## **6. Pakistan's SDG Progress**

- Pakistan ranked **129/165** (moderate progress)

### **Strong Areas:**

- Poverty reduction
- Health improvements

### **Weak Areas:**

- Education
- Climate action
- Governance

**Key Line:** Progress is uneven due to implementation gaps

## **7. Institutional Framework**

### **Key Bodies:**

- Prime Minister SDG Unit
- Parliamentary SDG Unit
- Planning Commission

### **Supporting:**

- Ministries + NGOs

**Example:** NGOs working in rural development

## **8. Role of CPEC**

### **Contributions:**

- Infrastructure
- Energy
- Jobs
- Poverty reduction

### **Critical View:**

- Debt concerns
- Environmental issues

## **9. Challenges**

- Economic crisis
- Political instability
- Weak institutions
- Lack of awareness
- Corruption
- Climate disasters

## **10. Recommendations**

- Economic reforms
- Better governance
- Increase funding
- Awareness campaigns
- Technology use
- International cooperation

## **Conclusion**

- SDGs are essential for Pakistan's future development
- Progress is mixed and uneven

## **Final Analytical Line:**

*“SDGs in Pakistan are limited not by vision but by governance, capacity, and resources.”*

# **Hybrid Warfare & Its Impacts on Pakistan**

## **1. Introduction**

- Warfare has transformed from traditional (kinetic) war → modern multidimensional conflict.
- Due to nuclear deterrence, especially after Cold War, direct wars between major states have become less likely.
- States now prefer Hybrid Warfare (HW) to achieve objectives without open confrontation.

**Key Concept:** Hybrid warfare operates in the “grey zone” (between war and peace).

- Pakistan, due to its:
  - Strategic location
  - Nuclear capability
  - Role in War on Terror

has become a primary target of hybrid strategies.

## **2. Meaning / Definition**

- Hybrid warfare is a synchronized use of multiple tools of power:
  - Military
  - Political
  - Economic
  - Informational
  - Cyber

### **Core Concept:**

Hybrid warfare = “Smart war using all possible tools to weaken an opponent internally”

### **Theoretical Insight:**

- Based on idea of “Total War” + “Psychological Warfare”
- Focus shifts from:
  - Territory → perception and control of narrative

**Example:** Creating distrust in elections through social media manipulation.

## **3. Key Features**

1. **Multi-Domain Operations**
  - Land, cyber, media, economy all combined
2. **Ambiguity**
  - Hard to identify attacker
3. **Low Intensity but Continuous**
  - No clear start or end
4. **Use of Technology**

- Social media, AI, cyber tools
- 5. **Targeting Society**
  - Focus on:
    - Public opinion
    - Youth
    - Institutions

#### **4. Historical Evolution**

##### **A. Sun Tzu**

- “The supreme art of war is to subdue the enemy without fighting”  
Basis of hybrid warfare

##### **B. Carl von Clausewitz**

- War = continuation of politics  
Hybrid warfare blends politics + war

##### **C. Cold War Era**

- Indirect conflict using:
  - Proxy wars
  - Economic aid
  - Ideology

Example: USA vs USSR influence competition

##### **D. Post-9/11 Era**

- Rise of:
  - Terrorism
  - Cyber warfare
  - Information warfare

#### **5. Modern Hybrid Warfare (21st Century)**

##### **Key Drivers:**

- Globalization
- Internet & social media
- Weak governance in developing states

##### **Role of Non-State Actors:**

- Al-Qaeda
- ISIS
- Tehrik-i-Taliban Pakistan

These groups use:

- Violence + propaganda + recruitment

## **6. State vs Non-State Hybrid Warfare**

### **A. Non-State Actors**

- Terrorism
- Guerrilla warfare
- Radicalization

### **B. State Actors**

- Cyber attacks
- Economic pressure
- Diplomatic isolation
- Media manipulation

### **Key Difference:**

State actors operate covertly, non-state actors operate violently and openly

## **7. Case Studies**

### **7.1 Annexation of Crimea**

- Russia used:
  - Cyber warfare
  - Media propaganda
  - Proxy forces

Result: Territory captured without full-scale war

### **7.2 Iran's Strategy in Middle East**

- Use of proxies:
  - Hezbollah
  - Houthis

Goal: Expand regional influence

### **7.3 India-Pakistan Context**

- Allegations of:
  - Information warfare
  - Diplomatic pressure

Example: Narrative-building in global media

## **8. Elements (Modes) of Hybrid Warfare**

### **A. Physical / Kinetic Modes**

### **1. Conventional Warfare**

- Direct military conflict

### **2. Asymmetric Warfare**

- Weak vs strong

### **3. Terrorism**

- Violence to create fear

### **4. Organized Crime**

- Drug trafficking, smuggling

Funds hybrid activities

## **B. Non-Kinetic / Conceptual Modes**

### **1. Political Warfare**

- Election interference

### **2. Economic Warfare**

- Sanctions, trade restrictions

### **3. Information Warfare**

- Fake news, propaganda

**Key Concept:** “Information is a weapon”

### **4. Cyber Warfare**

- Hacking systems

Example: Attacks on banks or government data

### **5. Diplomatic Warfare**

- Isolation in global forums

## **9. Hybrid Warfare Against Pakistan**

**Why Pakistan?**

- Strategic geography
- Nuclear power

- Role in War on Terror
- Importance of China Pakistan Economic Corridor

### **Main Threat Areas:**

- Political instability
- Ethnic tensions
- Economic vulnerability
- Media manipulation

## **10. Impacts on Pakistan**

### **A. Social Polarization**

#### **1. Ethnic Divisions**

- Balochistan issues

#### **2. Sectarian Conflict**

- Sunni vs Shia tensions

#### **3. Ideological Divide**

- Liberal vs conservative narratives

**Impact:** Weak national unity

### **B. Information Warfare & Media Manipulation**

- Fake news campaigns

Example: Anti-state narratives on social media

### **C. Anti-CPEC Campaigns**

- Propaganda against development

Goal: Reduce foreign investment

### **D. Economic Impact**

- Investor confidence decreases
- Currency instability

### **E. Terrorism & Security Threats**

- Attacks to destabilize

### **F. Diplomatic Pressure**

- Attempts to isolate Pakistan globally

## **11. Why Pakistan is Vulnerable?**

### **Internal Weaknesses:**

- Weak governance
- Political instability
- Economic crisis
- Social divisions

### **Key Concept:**

Hybrid warfare succeeds where internal weaknesses already exist

## **12. Counter Strategies / Solutions**

### **1. National Integration**

- Promote unity

### **2. Economic Stability**

- Strong economy reduces vulnerability

### **3. Media Literacy**

- Educate public against fake news

### **4. Cyber Security**

- Protect digital systems

### **5. Institutional Strengthening**

- Improve governance

### **6. Whole-of-Nation Approach**

- Cooperation between:
  - State
  - Military
  - Society

### **7. Strategic Communication**

- Build positive global narrative

## **Conclusion**

- Hybrid warfare is:

- Invisible
  - Continuous
  - Multi-dimensional
- It targets:
  - Minds
  - Society
  - Economy

*“Hybrid warfare is a battle for perception and control rather than territory.”*